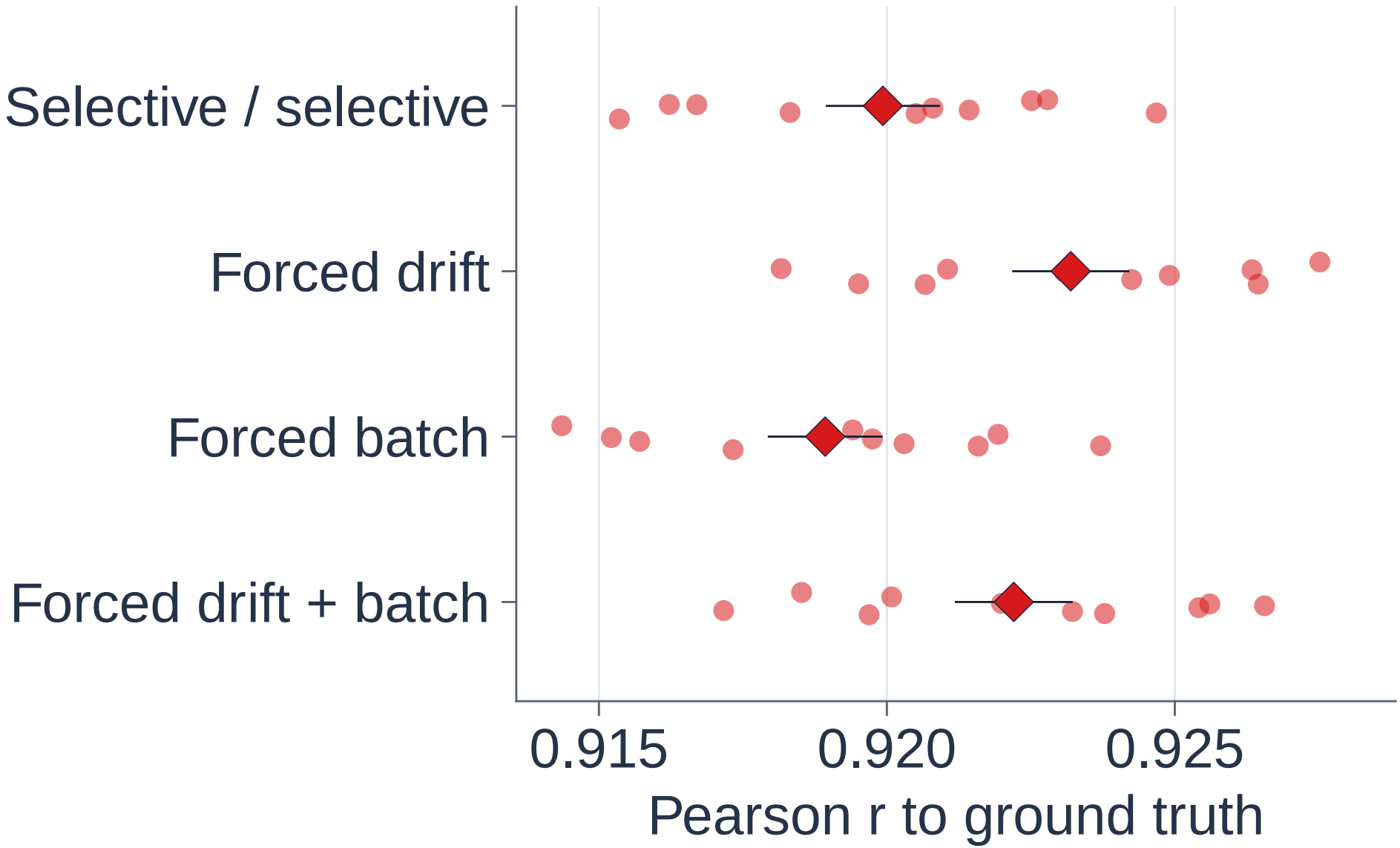F Simulation sample correlation